

Comprehensive Financial Mathematics & Formula Guide

Valuation, Capital Budgeting & Working Capital

Topic 1: Valuation Formulas

1. Free Cash Flow to Firm (FCFF)

The actual cash generated by the business available to all investors.

$$FCFF = EBIT \times (1 - t) + D\&A - \text{CapEx} - \Delta WC \quad (1)$$

Breakdown:

- *EBIT*: Earnings Before Interest and Taxes (Operating Profit).
- *t*: Corporate Tax Rate.
- *D&A*: Depreciation and Amortization (non-cash expenses added back).
- *CapEx*: Capital Expenditures (money spent on fixed assets).
- *ΔWC*: Change in Net Working Capital.

2. Capital Asset Pricing Model (CAPM)

Calculates the Cost of Equity (K_e).

$$K_e = R_f + \beta \times (R_m - R_f) \quad (2)$$

Breakdown:

- R_f : Risk-free rate (e.g., yield on a 10-year government bond).
- β : Beta (measure of the stock's volatility relative to the market).
- R_m : Expected market return.
- $(R_m - R_f)$: Market risk premium.

3. Weighted Average Cost of Capital (WACC)

The overall discount rate representing the blended cost of debt and equity.

$$WACC = K_e \left(\frac{E}{V} \right) + K_d \left(\frac{D}{V} \right) (1 - t) \quad (3)$$

Breakdown:

- K_e : Cost of Equity.
- K_d : Pre-tax Cost of Debt.
- E : Market Value of Equity.
- D : Market Value of Debt.
- V : Total Market Value ($E + D$).
- $(1 - t)$: Tax shield on debt (interest is tax-deductible).

4. Terminal Value (Gordon Growth Model)

The value of all cash flows beyond the projection period.

$$TV = \frac{FCF_n \times (1 + g)}{WACC - g} \quad (4)$$

Breakdown:

- FCF_n : Free Cash Flow in the final projected year.
- g : Perpetual growth rate (must be less than WACC).
- $WACC$: Discount rate.

Topic 2: Capital Budgeting Formulas

1. Net Present Value (NPV)

Measures the absolute dollar value created by a project.

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1 + r)^t} - C_0 \quad (5)$$

Breakdown:

- CF_t : Expected cash flow in period t .
- r : Discount rate (usually WACC).
- n : Total number of periods.
- C_0 : Initial Investment outlay at Time 0.

2. Internal Rate of Return (IRR)

The exact percentage return that forces NPV to equal exactly zero.

$$0 = \sum_{t=1}^n \frac{CF_t}{(1 + IRR)^t} - C_0 \quad (6)$$

Breakdown:

- *IRR*: The break-even discount rate. If $IRR > WACC$, accept the project.

3. Payback Period

Time required to recover the initial investment.

$$PBP = Y + \frac{\text{Unrecovered Cost at Start of Year } (Y + 1)}{\text{Cash Flow During Year } (Y + 1)} \quad (7)$$

Breakdown:

- *Y*: The last period with a negative cumulative cash flow.

Topic 3: Working Capital Management Formulas

1. Net Working Capital (NWC)

$$NWC = \text{Current Assets} - \text{Current Liabilities} \quad (8)$$

2. The Balance Sheet Cash Identity

Isolates the firm's liquid cash based on all other balance sheet components.

$$\text{Cash} = (LTD + \text{Equity}) - NWC_{excl.cash} - FA \quad (9)$$

Breakdown:

- *LTD*: Long-Term Debt.
- *Equity*: Shareholders' Equity.
- $NWC_{excl.cash}$: Net Working Capital minus the Cash balance.
- *FA*: Fixed Assets (Net).

Practice Problems

Problem 1 (Valuation):

Company XYZ has an EBIT of \$500,000. The corporate tax rate is 25%. Depreciation for the year is \$50,000. The company spent \$120,000 on new machinery (CapEx) and its Working Capital increased by \$30,000. Calculate the Free Cash Flow to the Firm (FCFF).

Problem 2 (Capital Budgeting):

Project Alpha requires an initial investment (C_0) of \$200,000. The firm's WACC is 12%. The projected after-tax cash flows are as follows:

- Year 1: \$60,000
- Year 2: \$80,000
- Year 3: \$90,000
- Year 4: \$70,000

Calculate the Net Present Value (NPV) of the project. Should the project be accepted?

Problem 3 (Working Capital):

McConnell Corp. has the following financial data: Book value of Equity = \$14,735; Long-term Debt = \$8,300; Net Working Capital (excluding cash) = \$2,850; Fixed Assets = \$18,440. Calculate exactly how much cash the company currently holds.

Solutions to Practice Problems

Solution 1 (Valuation):

$$\begin{aligned}
 FCFF &= [EBIT \times (1 - t)] + D\&A - \text{CapEx} - \Delta WC \\
 FCFF &= [\$500,000 \times (1 - 0.25)] + \$50,000 - \$120,000 - \$30,000 \\
 FCFF &= \$375,000 + \$50,000 - \$120,000 - \$30,000 \\
 \mathbf{FCFF} &= \mathbf{\$275,000}
 \end{aligned}$$

Solution 2 (Capital Budgeting):

$$\begin{aligned}
 NPV &= \frac{60,000}{(1.12)^1} + \frac{80,000}{(1.12)^2} + \frac{90,000}{(1.12)^3} + \frac{70,000}{(1.12)^4} - 200,000 \\
 NPV &= 53,571 + 63,776 + 64,066 + 44,484 - 200,000 \\
 NPV &= 225,897 - 200,000 \\
 \mathbf{NPV} &= \mathbf{\$25,897}
 \end{aligned}$$

Conclusion: Yes, accept the project because NPV ≥ 0 .

Solution 3 (Working Capital):

$$Cash = (LTD + Equity) - NWC_{excl.cash} - FA$$

$$Cash = (\$8,300 + \$14,735) - \$2,850 - \$18,440$$

$$Cash = \$23,035 - (\$21,290)$$

$$\mathbf{Cash = \$1,745}$$